MILITARY SPECIFICATION SHEET

GUIDANCE CONTROL SECTION, GUIDED MISSILE

RECEIVER SUBASSEMBLY

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the receiver subassembly described herein shall consist of this specification and the latest issue of MIL-G-85742.

1. SCOPE

1.1 Scope. This specification sheet establishes acceptance and test requirements for the receiver subassembly and related subassemblies described herein, used in the Guidance Control Section (GCS) of the SIDEARM Missile.

2. APPLICABLE DOCUMENTS

2.1 Government drawings. The following drawings, in addition to the applicable documents listed in MIL-G-85742, are applicable to this specification sheet.

Naval Air Systems Command
(Code Ident 30003)

1466AS1290	Local Oscillator.
1466AS1300	Receiver Section.
1466AS1312	Receiver Subassembly.
1466AS1314	Seeker Positioner Assembly.
1466AS1316	Loop Integrator.
1466As1318	Video Sample and Hold Assembly.
1466AS1320	IF Amplifier Assembly.
1466AS1348	Filter/Mixer Assembly

FSC 1420

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3. REQUIREMENTS

3.1 Item description. The receiver subassembly is comprised of the following subassemblies which form a part of the SIDEARM GCS receiver section, Drawing 1466AS1300. This assembly conditions electrical signals provided by the GCS seeker section, and outputs signals to the GCS servo section,

a. Receiver Subassembly (Drawing 1466AS1312).

b. Filter/Mixer Assembly (Drawing 1466AS1348).

c. IF Amplifier Assembly (Drawing 1466AS1320).

d. Local Oscillator (Drawing 1466AS1290).

e. Video Sample and Hold Assembly (Drawing 1466AS1318).

f. Loop Integrator (Drawing 1466AS1316).

g. Seeker Positioner Assembly (Drawing 1466AS1314).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Lock-on sensitivity and dynamic range. With the radio frequency (RF) attenuator enabled, the receiver subassembly shall have the capability to acquire and track targets over the frequency range of A to G whose power varies over at least the range from 2C to 1C. With the attenuator free to operate as directed by the automatic gain control (AGC) (no cage signal), the receiver subassembly shall have the capability to acquire and track targets over the frequency range A to G whose power level covers the range of at least 4B to 1D. Upon acquisition of targets over the power levels described, the receiver subassembly shall continue to track these targets to power levels of 1C + 10 decibels (dB) or greater. The RF attenuator shall be enabled to 43 ± 6 dB above the lower threshold signal level, provide 20 + 6 dB of attenuation and provide 8 ± 6 dB of hysteresis.

3.2.1.2 Target acquisition. The receiver subassembly shall provide the capability to scan the entire frequency band of A to G in a sawtooth manner. The scan time shall be T1 with a reset time of T2.

3.2.1.3 Target reacquisition. Upon loss of the target for T3, the receiver subassembly shall attempt to acquire a target by sweeping upward in frequency.

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3.2.1.4 Target reselect. If the receiver subassembly is tracking a target and the recage signal is applied, the receiver shall break lock and attempt to acquire a target at a frequency H above the original target frequency. If a target at a higher frequency is not present, the receiver shall return to the normal acquisition mode.

3.2.1,5 Video output pulse amplitude. The video output pulse amplitude shall be 3.0 ± 0.6 volts (V) for receiver input power levels of 10 dB above lower acquisition threshold to 0 decibels referenced to 1 milliwatt (dBm). This requirement shall be met for signal conditions defined in the Appendix. For a given receiver, the total variation over this power level shall not exceed 0.5 V.

3.2.1.6 Pilot tone. When the receiver threshold has been exceeded, the pilot tone will be the pulse repetition frequency (PRF) of the incoming signal. The pilot tone signal will be a pulse 100 ± 15 microseconds wide, 3.8 ± 0.2 volts peak-to-peak (Vp-p) at the PRF of the incoming signal.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Each receiver subassembly shall meet the requirements of 3.2 when inspected and tested as specified in Drawing 1466AS1312 and the following paragraphs. Failure to meet any of the requirements shall be cause for rejection of the receiver subassembly.

4.1.1 Test provisions and conditions. Testing shall be performed under the following environmental conditions.

a. Temperature (ambient): 23 ± 5 degrees Celsius.

b. Altitude: Elevation of test facility.

c. Relative humidity: 95 percent or less.

4.1.2 Test equipment accuracy. Unless otherwise specified, test equipment accuracy shall be as specified in MIL-G-85742.

4.1.3 Coded test parameters. Unless otherwise specified, test equipment accuracy shall be as specified in the Appendix.

4.2 Quality conformance inspections.

4.2.1 Test methods. Details of methods and procedures shall be in accordance with the test procedure.

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4.2.1.1 System current. Upon initial powerup of the receiver subassembly at the subassembly test station with no RF input, the receiver subassembly shall draw the following supply currents:

a. +15 V plus +15 V/L.0. cycling between 0.9 amperes (A) and 1.8 A (due to oscillator tuning).

b. -15 V, 100 ± 50 milliamperes (mA).

c. +5 V, 250 ± 100 mA.

d. +27 V, not greater than 100 mA.

e. -27 V, not greater than 100 mA.

4.2.1.2 Lock on sensitivity and dynamic range. The receiver is caged when 15 V in series with a 2 kilohm resistor is applied to the cage enable input. The receiver is uncaged when this 15 V is removed. Connect a RF pulse signal source to the RF input. Measure the caged and uncaged threshold at frequencies A and G, pulsed by PRF number (no.) 1, pulse width (PW) no. 5. Measure the caged and uncaged threshold at frequencies B through F, pulsed by PRF no. 2, PW no. 5. Threshold occurs when (1) frequency lock on-telemetry (TLMY) changes state from 0 +0.3, -0.0 V to 4.1 ± 0.3 V;(2) the automatic frequency control (NC) voltage-TLMY stops sweeping; and (3) a stable, negative going pulse is present at Video test point (TP). The caged lower threshold shall be less than 2C. The uncaged lower threshold shall be less than 4B. The difference between the caged and uncaged thresholds shall be 20 ± 6 dB. With the receiver uncaged, measure the power level at which the RF attenuator is enabled at frequencies A to G. The RF attenuator is enabled when Attenuator Control-TLMY changes state from 0 +0.2, -0.0 V to 14.3 ± 0.3V. The RF attenuator shall enable 43 ±5 dB above uncaged lower threshold. Once the RF attenuator is enabled, the RF attenuator hysteresis is obtained by reducing the input power until the RF attenuator is disabled. The RF attenuator hysteresis is the difference between the power levels of enablement and disablement. The RF attenuator hysteresis shall be 8 ± 6 dB.

4.2.1.3 Frequency scan. With no RF input, measure the rise and fall times of the sawtooth waveform on AFC voltage-TLMY. The rise time shall be 1.2 ± 0.1 sec. The fall time shall be not greater than 0.1 sec. The voltage of the waveform shall be not less than 0.0 ± 0.5 V. The voltage of the waveform shall be not greater than 10.0 ± 0.5 V.

4.2.1.4 Target reacquisition. While tracking a target at frequency D, pulsed by PRF no. 1, PW no. 5, remove the target for 100 ms. The AFC voltage-TLMY shall begin sweeping upward to reacquire. The receiver shall reacquire the target in one cycle of the AFC voltage-TLMY waveform.

4.2.1.5 Target reselect. While tracking a target frequency D pulsed by PRF no. 1, PW no. 5 at power level 4B + 5 dB, in the uncaged mode and in the presence of a target at frequency D + H pulsed by PRF no. 1, PW no. 7 at power level 2C + 5 dB, cage the receiver subassembly. The receiver shall

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break lock and acquire the target at D + H as evidenced by an increase in pulse width at Video TP. Remove the RF signal at D + H. With the receiver subassembly tracking the target at frequency D uncage and cage the receiver. The receiver shall break lock and reacquire the target at frequency D.

4.2.1.6 Video output pulse amplitude. While tracking a target in the uncaged mode at frequency D pulsed by PRF no. 1, PW no. 5 at the power level 10 dB above the uncaged lower threshold at frequency D, the video pulse amplitude at Video TP shall be 3.0 ± 0.6 V. Increase the RF power to 0 3 dbm. The pulse amplitude at Video TP shall be 3.0 ± 0.6 V. The difference between the two pulse amplitudes shall not exceed 0.5 V.

4.2.1.7 Pilot tone. While tracking a target at frequency D pulsed at PRF no. 1, PW no. 5, the pilot headtone output shall be a pulse 100 ± 15 µs wide, 3.8 ± 0 .2 Vp-p at PRF no. 1.

Preparing activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-85742/2(AS)	Receiver Subassembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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